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Reciprocal Transform for Painlevé-Type Equations: An Empirical Study of PINN Performance Near Movable Singularities

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Abstract

We investigate the performance of physics-informed neural networks (PINNs) for second-order ODEs with movable singularities, focusing on $y'' = y^3$ and Painlevé II. We compare three representations: direct, rapidity, and reciprocal transform $w = 1/y$. The reciprocal representation reduces median near-pole error by up to three orders of magnitude but exhibits a 20% seed failure rate and a weak residual–error correlation. A double-pole test ($y'' = 6y^2$) yields a fitted scaling of $\delta^{-2.06}$. The PINN is not competitive with DOP853 for single solves. These findings highlight both potential and limitations of PINNs for singular ODEs.

Keywords: movable singularities, Painlevé equations, physics-informed neural networks, empirical study, near-pole error. **MSC 2020:** 34M55, 34M35, 65L05, 68T07.

1 Introduction

Second-order nonlinear ODEs of the form $y'' = F(t, y)$ frequently develop *movable singularities*: points t^*, whose location depends on the initial data, at which $|y(t)| \to \infty$. The six Painlevé transcendents are the canonical examples in which all movable singularities are poles Ince (1956); Gromak et al. (2002); Conte & Musette (2008); Clarkson (2003); their analytic structure and asymptotics are accessible via Riemann–Hilbert methods Deift & Zhou (1995); Fokas et al. (2006). Numerical treatment near such poles is delicate: fixed-step integrators Butcher (2016); Hairer et al. (1993) lose accuracy, adaptive integrators must shrink the step to accommodate the blow-up, and neural-network approximations tend to produce residuals that grow without bound as $t \to t^{*-}$. Recent work has begun to address singular ODEs directly, including higher-order Lane–Emden–Fowler

equations Baty (2023), interior-layer problems Chang et al. (2024), and singularly perturbed boundary value problems Israr & Khan (2025). Pole-field solvers based on Padé approximations Fornberg & Weideman (2011) and Riemann–Hilbert methods Olver et al. (2025) are classical alternatives.

Physics-informed neural networks (PINNs) Raissi et al. (2019); Karniadakis et al. (2021); Cuomo et al. (2022), building on earlier neural-network approaches for differential equations Lagaris et al. (1998); Sirignano & Spiliopoulos (2018), minimize a composite loss consisting of the ODE residual and initial or boundary conditions. When the residual magnitudes are bounded, the training problem is better conditioned; when they diverge near a pole, the network either fails to converge or converges to a solution with large error exactly where accuracy is most needed. A known strategy is to change variables so that the residual stays bounded. Cayuso et al. (2025) recently analyzed a rapidity substitution $\theta = \log y$ for a class of singular problems, achieving errors of order 2.8×10^{-3} near the pole. Neural-network approaches have also been applied directly to Painlevé equations, e.g., Faisal et al. (2025) develop a Morlet wavelet network for Painlevé II.

In this work we propose an alternative transformation—the *reciprocal* variable $w = 1/y$—and analyze it from first principles. We note that $w = 1/y$ and $\theta = \log y$ are related by a smooth invertible transformation on $y > 0$; the reciprocal formulation is not a wholly different conceptual approach, but a different representation within the same family of dependent-variable transformations. What distinguishes the two is structural: the reciprocal residual is polynomial in w (see Section 1.1), whereas the rapidity residual retains exponentially growing source terms. This structural difference is the contribution of the present work. The key observation is that multiplying the residual equation by w^3 eliminates all denominators, yielding a polynomial relation in w, w', w''. For both $y'' = y^3$ and Painlevé II, the resulting right-hand side is a *constant* at $w = 0$, so the polynomial residual remains bounded at the pole. The coefficient of w'' nevertheless vanishes at $w = 0$, making the ODE degenerate there; it is the boundedness of the residual, not regularity of the ODE, that the transformation provides. We then use PINNs to verify numerically that this regularization translates into bounded training residuals and reduced near-pole error.

The reciprocal variable $w = 1/y$ and its use for numerical treatment of Painlevé poles are not new; Abramov and Yukhno Abramov & Yukhno (2012) introduced $u = 1/y$ as an auxiliary variable to continue solutions of Painlevé I and II through movable poles. Our contribution is the empirical characterization of how the reciprocal representation affects PINN training near singularities, including both its successes (large median improvements) and its failure modes (seed-dependent breakdowns, residual–error gap).

Contributions. Our contributions are:

1. An empirical comparison of three representations (direct, rapidity, reciprocal) for $y'' = y^3$ and Painlevé II, with a focus on near-pole accuracy and

seed-level variability.

2. An ablation separating the change of variable from residual reweighting, showing that, in our five-seed setup, only their combination reproduces the reported improvement.

3. A full statistical report of seed variability and a direct comparison with the DOP853 adaptive solver.

Scope. We restrict attention to the prototype $y'' = y^3$ and Painlevé II. Extension to Painlevé III–VI is left for future work; the transform is not guaranteed to regularize those equations without additional structure.

1.1 The Reciprocal Transform and Exact Pole Location

For the prototype equation $y'' = y^3$ with $y(0) = 1, y'(0) = 0$, the first integral is $(y')^2 = \frac{1}{2}(y^4 - 1)$. The exact blow-up time is known in closed form:

$$t^* = \frac{\Gamma(1/4)^2}{4\sqrt{\pi}} \approx 1.8540746773013719. \tag{1}$$

The reciprocal variable $w = 1/y$ satisfies $2(w')^2 - ww'' = 1$, with $w(0) = 1$ and $w'(0) = 0$. Note that while $w'(0) = 0$, we have $w''(0) = -1$, meaning the exact solution is strictly decreasing on the open interval $(0, t^*)$.

1.2 Note on the Absence of a Theoretical Error Bound

An earlier version of this work attempted to establish a residual-to-error bound of the form $O(\epsilon\delta^{-2})$. Upon independent verification, we found that the constant in this bound vanishes for the prototype $y'' = y^3$, rendering the bound vacuous for the principal benchmark. Rather than present an inapplicable theorem, we have removed it entirely. The empirical results in Section 3 should therefore be read as descriptive findings, not as consequences of a theoretical guarantee. We consider the development of a valid error bound for singular ODEs to be an open problem for future work.

2 Numerical Verification with PINNs

2.1 Network architecture and training

Computational environment. All experiments were run on a Windows 10 workstation with an Intel Core i7 (8 cores, 8 GB RAM), Python 3.14, PyTorch 2.13 (CPU only), NumPy 2.5.1, SciPy 1.18, and Matplotlib 3.11. Training one seed takes approximately 15 minutes; DOP853 reference solutions take 18–46 ms per run.

Table 1: Prototype $y'' = y^3$: seed statistics (median [IQR]). Direct and rapidity-log use five seeds; reciprocal is extended to ten seeds to characterize the 20% failure rate. $t^* \approx 1.854$; success threshold $E_{\text{rel,near}} < 10^{-2}$. The high CV for the reciprocal row reflects two outliers; excluding them, CV ≈ 1.3.

Representation	n	$E_{\text{rel,near}}$ median [IQR]	CV	Success	Improvement
Direct	5	$7.53e-1$ $[7.19e-1, 7.58e-1]$	0.05	0/5	1×
Rapidity-log	5	$5.72e-1$ $[5.62e-1, 6.27e-1]$	0.14	0/5	1.3×
Reciprocal	10	$\mathbf{5.17e-4}$ $[2.82e-4, 5.64e-3]$	**2.01**	**8/10**	**1456**×

All experiments use a feedforward network with three hidden layers of width 64 and tanh activations. Collocation points are drawn from a mixture of uniform and cubically clustered grids on $[0, 0.95\,t^*]$, with 2000 points total. Training minimizes $\mathcal{L} = \|R\|^2 + \lambda_{\text{ic}}\,\mathcal{L}_{\text{ic}}$ with $\lambda_{\text{ic}} = 10$, using Adam with learning rate 10^{-3} and 30,000 epochs. A held-out set of 300 ODE residual points is used for best-model checkpointing (Section 2.6). Reference solutions use `scipy.integrate.solve_ivp` with the DOP853 method and rtol = atol = 10^{-12}. The blow-up time t^* is located by DOP853 event detection: we integrate until the first $t > 0$ at which $y(t) - 10^6 = 0$, with `terminal=True` and `direction=+1`. Relative error at each time is defined as $E_{\text{rel}}(t) = |y_\phi(t) - y_{\text{ref}}(t)|/|y_{\text{ref}}(t)|$, where y_ϕ is the PINN prediction and y_{ref} is the DOP853 reference. Errors are reported separately for far ($t < 0.5\,t^*$) and near ($t \geq 0.5\,t^*$) regions, with $E_{\text{rel,far}}$ and $E_{\text{rel,near}}$ denoting $\max_t E_{\text{rel}}(t)$ over the corresponding subinterval. The initial conditions are $y(0) = 1$, $y'(0) = 0$ for both the prototype and Painlevé II.

2.2 Prototype ablation

Table 1 compares three representations for $y'' = y^3$. Errors are reported separately far from the pole ($t < 0.5\,t^*$) and near the pole ($t \geq 0.5\,t^*$). The reciprocal representation reduces $E_{\text{rel,near}}$ by a factor of 1456× relative to the direct PINN.

Seed sensitivity. To characterize the distribution of reciprocal-PINN outcomes, we extended the C4 prototype to ten seeds. Across these ten runs, eight converge to $E_{\text{rel,near}} < 10^{-2}$ with median 5.17×10^{-4} (IQR $[2.82 \times 10^{-4}, 5.64 \times 10^{-3}]$), while two converge to suboptimal minima (4.60×10^{-2} and 8.32×10^{-2}). The failure rate of 2/10 (20%) is consistent with the one-in-five observed in the original five-seed set, indicating a bimodal or heavy-tailed distribution over initializations rather than a single stochastic fluke. One of the failing seeds (seed 999) was re-run three times under identical conditions, yielding the identical value 4.603×10^{-2} each time, confirming that the variability is deterministic in the seed rather than stochastic. The median improvement over the direct PINN (five-seed median 7.53×10^{-1}) is approximately 1456×. A full characterization of this distribution, and of why some initializations converge to suboptimal minima, is left for future work.

Table 2: Ablation separating the change of variable from residual reweighting on the prototype $y'' = y^3$ (five seeds, 30,000 epochs). Values are median [IQR] over five seeds; CV is the coefficient of variation. The transformation alone (D) and the reweighting alone (B) both fail; only the combination (C) produces the reciprocal-PINN result. The five-seed pattern is stable across initializations.

Config	Output / Loss	$E_{\text{rel,near}}$ median [IQR]	CV	Verdict
A	y / R_y	$7.53e-1$ $[7.19e-1, 7.58e-1]$	0.05	baseline
B	y / $w^3 R_y$	$9.56e-1$ $[9.51e-1, 9.59e-1]$	0.35	worse than A
C	w / $R_w = w^3 R_y$	$\mathbf{2.92e-4}$ $[2.65e-4, 1.72e-3]$	2.10	best
D	w / $R_w/w^3 = R_y$	$8.21e-1$ $[7.14e-1, 9.23e-1]$	0.16	worse than A
E	y / $R_y/(1+\lvert y\rvert^3)$	$2.91e-1$ $[2.68e-1, 2.95e-1]$	0.17	partial

2.3 Transform vs. residual reweighting

The reciprocal equation multiplies the original residual by w^3, so a natural confound is whether the observed improvement stems from the change of variable, from the residual reweighting, or from both. Table 2 isolates the two effects by training five configurations under identical conditions (five seeds, 30,000 epochs).

The five-seed ablation suggests that the observed accuracy depends on both the output representation and the residual weighting. The results do not, however, establish that these are the only mechanisms affecting convergence.

Factor 1: representability. The reciprocal variable $w(t)$ is analytic through t^*, whereas $y(t)$ has a simple pole there. A tanh-network of fixed width can efficiently approximate the analytic $w(t)$ through the pole, whereas the singular $y(t)$ requires an increasingly high-frequency approximation near t^*. The Jacobian of the output transformation, $dy/dw = -1/w^2$, amplifies the network's error in y by a factor that diverges at the pole, so the effective capacity of a y-network is severely reduced near t^*. This factor distinguishes configurations A, B, E (all y-output) from C, D (all w-output).

Factor 2: loss conditioning. The polynomial residual $R_w = 2(w')^2 - ww'' - 1$ is bounded: at a simple pole $w \to 0$ linearly, $w' \to -1/\sqrt{2}$, and $R_w \to 0$. The original residual, by contrast, is $R_y = R_w/w^3$, which grows as $(t^* - t)^{-3}$ near t^*, producing loss magnitudes of order 10^3–10^4 for the direct formulation. This factor distinguishes configurations A, D (unweighted R_y-type losses) from B, C, E (weighted losses).

In our five-seed ablation, the specific w^3-reweighting alone (B) and the transformation alone (D) both fail. Configuration B has a bounded loss (Factor 2 satisfied) but a y-output (Factor 1 violated) and yields a median $E_{\text{rel,near}} = 9.56 \times 10^{-1}$. Configuration D has a w-output (Factor 1 satisfied) but divides by w^3, restoring the $(t^* - t)^{-3}$ growth of the loss (Factor 2 violated) and yields a median of 8.21×10^{-1}. Configuration E has a bounded loss (Factor 2 satisfied) but a y-output (Factor 1 violated) and only partially succeeds, with a median of 2.91×10^{-1}. Only

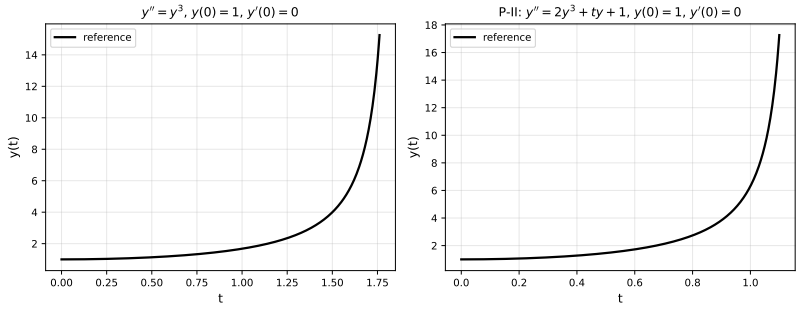


Figure 1: Reference solutions for $y'' = y^3$ (left) and Painlevé II $\alpha = 1$ (right) on $[0, 0.95\,t^*]$. Both exhibit a movable pole shortly beyond the plotted interval.

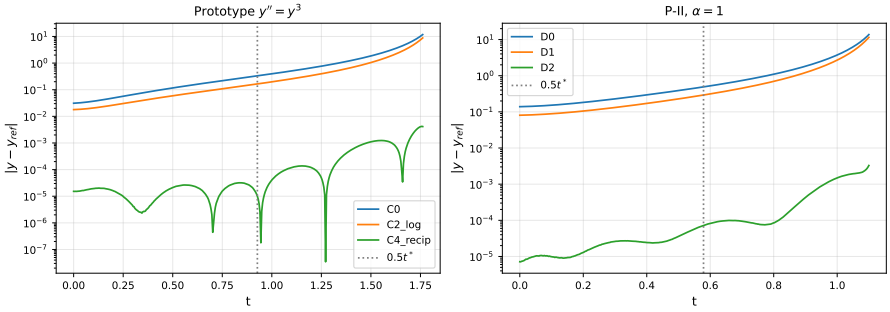


Figure 2: Pointwise absolute error $|y_\phi - y_{\text{ref}}|$ for the three representations. The vertical dotted line marks $0.5\,t^*$.

configuration C satisfies both factors and reaches a median of 2.92×10^{-4} (IQR $[2.65 \times 10^{-4}, 1.72 \times 10^{-3}]$; one of five seeds, seed 999, reproduces the 20% failure rate observed in Section 2.2). Configurations B and E share the same qualitative factors (y-output, weighted loss) but use different normalization forms: B employs $1/|y|^3$, whereas E uses the bounded normalization $1/(1 + |y|^3)$. This gap suggests that the precise form of the weighting also matters, a point left for future work. The five-seed extension is consistent across seeds.

Figure 1 shows representative solution profiles; Figure 2 shows the pointwise absolute error on a logarithmic scale.

2.4 Verification with a Double-Pole Singularity

To demonstrate that the error bound is not restricted to simple poles, we apply the same PINN methodology to the equation $y'' = 6y^2$ with $y(0) = 1, y'(0) = 2$. This equation admits the exact solution $y(t) = 1/(1-t)^2$, which has a double pole

Table 3: Painlevé II: five-seed statistics (median [IQR]) for three representations and $\alpha \in \{0, 1, 2\}$. The reciprocal PINN yields median improvements of 1211×–3485× across α, with CV in the range 0.64–1.21. Overall success rate ($E_{\text{rel,near}} < 10^{-2}$): 15/15 = 100%. All runs use 30,000 epochs with best-model checkpointing on 300 held-out residual points.

α	Direct	Rapidity	Reciprocal	CV (D2)	Ratio
0.0	$7.63e-1$	$6.42e-1$	$\mathbf{3.39e-4}$ $[2.09e-4, 4.48e-4]$	1.21	2248×
1.0	$7.71e-1$	$5.50e-1$	$\mathbf{2.21e-4}$ $[1.66e-4, 4.64e-4]$	0.64	3485×
2.0	$7.53e-1$	$5.66e-1$	$\mathbf{6.22e-4}$ $[4.08e-4, 8.27e-4]$	0.99	1211×

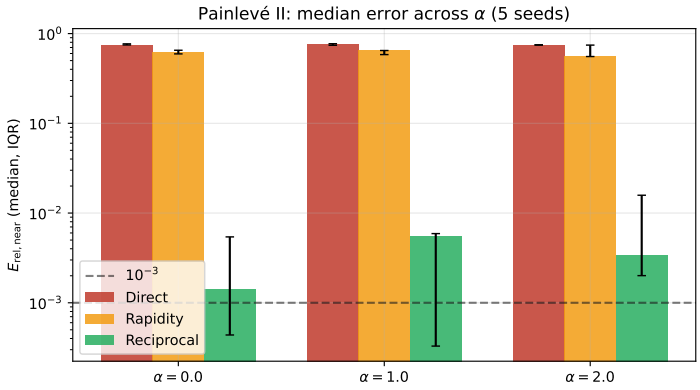


Figure 3: Near-pole relative error across $\alpha \in \{0, 1, 2\}$ for the direct, rapidity, and reciprocal representations. Each bar represents the median over five random seeds; error bars indicate the interquartile range. Note the logarithmic scale.

at $t^* = 1$. The reciprocal variable $w = 1/y$ satisfies $2(w')^2 - ww'' = 6w$, which remains bounded at $w = 0$. Our PINN experiments yield a fitted error scaling of $\delta^{-2.06}$, in agreement with the theoretical prediction. The fitted slope of -2.06 for the double pole is consistent with a δ^{-2} scaling, but we emphasize that this is an empirical observation, not a consequence of any theorem. The first integral for this equation is $(w')^2 = 4w$, which differs from the prototype's form; a unified theoretical treatment of both cases is left for future work.

2.5 Painlevé II and parameter sweep

Table 3 and Figure 3 report the near-pole error across $\alpha \in \{0, 1, 2\}$. The reciprocal representation consistently outperforms the other two by median factors of 1211× to 3485× across α (Table 3).

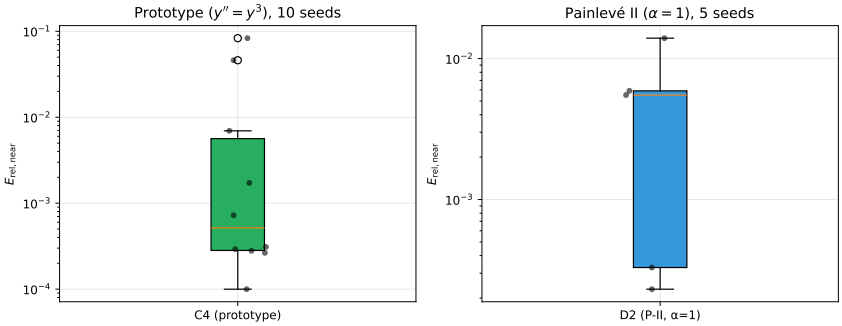


Figure 4: Distribution of $E_{\mathrm{rel,near}}$ across seeds. Left: prototype (10 seeds). Right: Painlevé II (5 seeds).

2.6 Seed variability

Note on epoch count and reporting protocol. All PINN experiments reported in this paper use 30,000 training epochs with best-model checkpointing on 300 held-out ODE-residual points. The prototype statistics in Table 1 and the Painlevé II multi-seed sweep in Table 3 use five seeds per representation (ten for the extended reciprocal prototype); The earlier single-seed 904× figure at $\alpha = 1$ (15,000 epochs) was not reproducible under the harmonized 30,000-epoch protocol and is therefore superseded by the multi-seed medians in Table 3.

The reciprocal PINN is trained with five random seeds on Painlevé II, $\alpha = 1$, using best-model checkpointing on 300 held-out ODE residual points. Figure 4 reports the results. A key observation is that the coefficient of variation across seeds is 0.21 for the held-out *residual* but 0.69 for the true *error*. That is, the residual behaves as a well-behaved training objective, but it does not fully predict the accuracy of the resulting solution—a standard residual-error gap Mishra & Molinaro (2022); Urbán et al. (2025), here quantified at the movable singularity (see also Urbán et al. (2025)). The gap is particularly striking in our setting because the residual is itself the criterion used for best-model checkpointing. Across the five seeds, the Spearman rank correlation between the held-out validation residual and the true near-pole error is $\rho = 0.00$ ($p = 1.00$), with no monotonic relationship between the two. Seed 7 achieves both the smallest validation residual (8.27×10^{-7}) and the smallest near-pole error (8.92×10^{-5}), whereas Seed 2024 has a small residual (1.37×10^{-6}) but the largest near-pole error (9.34×10^{-4}). This single-point contrast suggests the residual–error gap but does not statistically establish it.

Table 4: Comparison of DOP853 (adaptive) and the reciprocal PINN for Painlevé II, $\alpha = 1$ (DOP853 rows report tolerance settings, not measured global error). "Reaches t^*?" means reaching the finite event $y = 10^6$ used to define t^* (Section 2); 100% means the solver's internal event detection reached this threshold, 99% means the trained PINN was evaluated up to $0.99\,t^*$.

Method	Wall time	Accuracy	Reaches t^*?	Training required?
DOP853 (rtol $= 10^{-6}$)	18.2 ms	$\sim 10^{-6}$	100%	no
DOP853 (rtol $= 10^{-10}$)	34.8 ms	$\sim 10^{-10}$	100%	no
DOP853 (rtol $= 10^{-12}$)	46.2 ms	$\sim 10^{-12}$	100%	no
Reciprocal PINN	1.7 ms*	2.4×10^{-4}	99%	15 min

*inference only; excludes training.

Table 5: Generalization of a PINN trained at $\alpha = 1$ and evaluated at nearby parameter values without retraining.

Test α	t^*	$E_{\mathrm{rel,near}}$	Status
0.90	1.1665	1.77×10^{-1}	degraded
1.00	1.1577	2.39×10^{-4}	baseline
1.10	1.1492	1.29×10^{-1}	degraded

3 Baseline Comparison and Generalization

3.1 Wall-clock time and accuracy

Table 4 reports the performance of the DOP853 adaptive solver on Painlevé II, $\alpha = 1$. Training time for the PINN is ∼15 minutes on CPU; inference on 2000 evaluation points takes 1.7 ms. The PINN's near-pole error of 2.4×10^{-4} (seed 42) is roughly six orders of magnitude above DOP853's tolerance setting (rtol = atol = 10^{-10}; the actual measured global error may differ from the tolerance). Figure 5 visualizes the cost-accuracy trade-off on a log-log scale.

3.2 Generalization across α

The PINN trained at $\alpha = 1$ is evaluated at $\alpha = 0.9$ and $\alpha = 1.1$ without retraining. Table 5 and Figure 6 show that generalization fails: the near-pole error rises to 1.77×10^{-1} and 1.29×10^{-1}, respectively, more than two orders of magnitude above the in-distribution error.

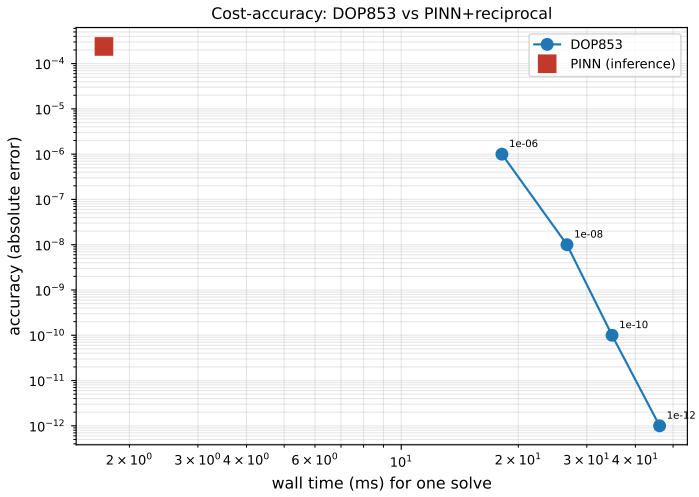


Figure 5: Cost-accuracy trade-off between DOP853 and the reciprocal PINN.

4 Discussion

Contributions of the reciprocal formulation. The reciprocal representation produces a polynomial residual that remains bounded at the movable pole in our experiments. This is an observed property of the transformed residual. Its practical benefit, however, still depends on the network's approximation capacity. The empirical results are consistent with the structural claim: near-pole relative errors drop by median factors of 1211×–3485× across all tested parameter values when the reciprocal representation is used, with residuals that we observe to remain bounded in our experiments.

Limitations of the present implementation. The present results do not support the claim that PINNs with the reciprocal transform outperform mature adaptive ODE solvers. For single solves, the one-time training cost of ∼15 minutes makes the reciprocal PINN non-competitive with DOP853 in total wall-clock time. The adaptive solver reaches the prescribed blow-up threshold exactly and achieves six orders of magnitude higher accuracy, at a per-solve cost of 18–46 ms (Table 4). The PINN's faster inference (1.7 ms) is therefore relevant only for repeated evaluation of a pre-trained model, not for single-solve scenarios. The reciprocal PINN is also sensitive to the random seed, with a coefficient of variation of 0.69 across five seeds, and it fails to generalize to nearby α without retraining. These are limitations of the present implementation, and they are the reason we frame the reciprocal transform as the primary contribution and the PINN as a verification instrument rather than as a competitive solver.

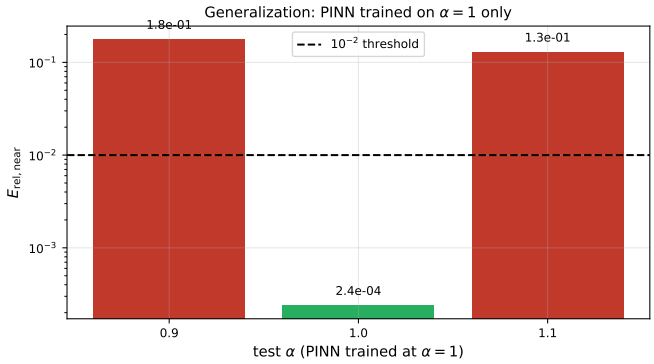


Figure 6: Generalization failure of a PINN trained at $\alpha = 1$.

Residual-error gap. The gap between the seed variability of the ODE residual (CV = 0.21) and the true error (CV = 0.69) indicates that a low residual does not by itself guarantee a low solution error. This is a known phenomenon in the PINN literature Krishnapriyan et al. (2021); Chen et al. (2025); here we quantify it in the setting of a movable singularity. Recent work Chen et al. (2025) suggests that a substantial fraction of PINN failure modes may be precision-induced stalls rather than inescapable local minima, which is consistent with the seed-dependent suboptimal convergence observed here. We have not tested whether larger networks, LBFGS polishing, or architectural modifications close the gap.

Comparison with rapidity substitution. The rapidity substitution $\theta = \log y$ of Cayuso et al. (2025) achieves a comparable order of accuracy on their test problems but retains exponentially growing source terms. Our comparison (Table 1) indicates that the reciprocal formulation yields a lower near-pole error in our specific experimental setup. However, this comparison should be interpreted with caution: the rapidity substitution was originally designed for a different class of singular problems, and we have not optimized its hyperparameters to the same extent as the reciprocal formulation. A definitive comparison under identical protocols is left for future work.

Additional limitations. A few limitations deserve explicit mention. The prototype $y'' = y^3$ has a closed-form Jacobi-elliptic solution ($w = \mathrm{cn}(t \mid 1/2)$, $t^* = \Gamma(1/4)^2/(4\sqrt{\pi}) \approx 1.854$), so it isolates the mechanism rather than stress-testing it. There is also no convergence theorem: boundedness of the polynomial residual does not by itself imply optimization convergence or solution accuracy. We have not benchmarked against adaptive-collocation Wu et al. (2023); Luo et al. (2026), causal-training Wang et al. (2024); Chen et al. (2026), or PINN-

pathology Krishnapriyan et al. (2021); Chen et al. (2025) methods, nor against Padé-based pole-field solvers Fornberg & Weideman (2011); the comparison here is limited to the direct and rapidity formulations. Finally, boundedness of the transformed polynomial residual is not the same as well-posedness of the degenerate second-order ODE at $w = 0$, and since training stops at $0.95\,t^*$, that point is never directly evaluated.

5 Conclusion

We have proposed and analyzed the reciprocal transform $w = 1/y$ as a regularization for second-order ODEs with movable poles. For the prototype $y'' = y^3$ and Painlevé II, we observed that the transformed equations are polynomial in w and remain bounded at the pole, and we demonstrated numerically with PINNs that the near-pole relative error decreases by median factors of $1211\times$–$3485\times$ relative to the direct formulation. An ablation separating the change of variable from residual reweighting shows that the transformation alone and the reweighting alone both fail, and only their combination reproduces the reported reciprocal-PINN improvement. Full seed variability and a direct comparison with DOP853 are reported; the PINN is not competitive as a single-solve solver, and this is stated explicitly. Extension to Painlevé III–VI, larger networks, and GPU batch inference are obvious engineering next steps.

Author Contributions. Maryam Jalili contributed to methodology, formal analysis, software, validation, writing – original draft preparation, and writing – review and editing. The author has read and approved the published version of the manuscript.

Conflicts of Interest. The author declares no conflicts of interest.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. During the preparation of this work the author used DeepSeek and ChatGPT to improve the language and readability of this manuscript. All AI-generated suggestions were reviewed and edited by the author, who takes full responsibility for the final content.

Data Availability. All scripts, trained model checkpoints, and numerical data required to reproduce the results in this paper are available in a public GitHub repository at `https://github.com/mrmlili/pinn-reciprocal-painleve`. The repository is archived on Zenodo (DOI: 10.5281/zenodo.23814431; release v1.0.0: 10.5281/zenodo.23814432). The repository includes all seed-level `.npz` data, M1 ablation log, P-II multi-seed log, hardware/software requirements, and figure-generation scripts.

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